

Implementing Artificial Intelligence and Centralized Data Systems in Albania: A Model for Improving the Tourist Experience and Optimizing Small "Rent a Car" Enterprises

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Abstract

This academic paper analyzes the impact of Artificial Intelligence (AI) and its interconnection with official state databases on the tourism sector in Albania. The main focus is on small and medium-sized enterprises (SMEs) in the "Rent a Car" segment. The article investigates how the integration of intelligent algorithms with the unified platforms of the Ministry of Tourism serves as an efficient mechanism for simplifying operational administrative procedures, preventive legal education for foreign tourists regarding the Road Code and the Albanian Penal Code, and the intelligent management of electric vehicle (EV) logistics. This study integrates a literature review, a mixed methodological framework, and a Python simulation model for the energy consumption of electric vehicles on Albanian roads.

Keywords: Artificial Intelligence, Smart Tourism, Rent a Car, Electric Vehicles, Albanian Penal Code, Sustainable Development.

1. Introduction and Background of the Problem

In recent years, the Republic of Albania has experienced an unprecedented influx of international visitors. This exponential growth has exposed pronounced informational and logistical asymmetries among small local economic operators, particularly in the vehicle rental (Rent a Car) sector. The main challenges include language barriers, delays in verifying foreign documentation, and ambiguities regarding the country's road and penal regulatory framework.

Through an analytical approach, this paper argues that the application of AI does not constitute merely a commercial advantage, but an infrastructural pillar that, when coordinated with the centralized policies of the Ministry of Tourism and Environment, maximizes the value of the tourist experience (Consumer Experience - CX).

2. Literature Review

The digital transformation of tourism through Artificial Intelligence has become a central field of study in the management of modern services. According to Florido-Benítez and del Alcázar Martínez (2024), the integration of generative AI and intelligent agents has fundamentally changed consumer interaction in smart tourist destinations by eliminating language and administrative barriers, while also enhancing tourism marketing, personalization, and operational efficiency.

In the context of transport and tourist mobility, the OECD (2024) emphasizes that small and medium-sized enterprises (SMEs), which dominate the tourism sector, often have limited scale-up and innovation capacity and can suffer from information asymmetry relative to larger operators. State intervention through open data infrastructures is seen as a means of providing a more level competitive playing field for these smaller operators.

Finally, Baden's (2025) integrative literature review of EV range anxiety confirms that this phenomenon extends beyond simple distance concerns to encompass technical, psychological, and rhetorical facets, and notes that in-vehicle systems providing real-time, context-specific guidance — including topography — can help drivers make more informed route-planning decisions and reduce uncertainty. This supports a shift from static maps to dynamic, AI-based prediction systems that calculate consumption based on terrain topography — a key element that this paper specifically addresses for Albanian territory.

3. Research Methodology

This study uses a Mixed Methods Approach divided into two main phases:

Qualitative and Analytical Phase: Secondary data from the Albanian legal framework (the Road Code and Article 291 of the Penal Code) and strategies published by the Ministry of Tourism and Environment were analyzed. This served to construct the decision tree of the AI algorithm regarding client filtering by nationality.

Quantitative and Simulation Phase: A mathematical simulation model (designed in Python) was built based on the physics equations of transport (air resistance, rolling resistance, and gravitational force as a function of road grade). This model simulates the battery consumption of an EV on two typical Albanian tourist routes with differing topography: Tirana - Saranda (via the Llogara Pass) and Tirana - Shkodra.

4. The Interconnection Between AI and the Ministry of Tourism's Database

The creation of a unified database by the Ministry of Tourism enables integration, via APIs, with small business systems. This integration eliminates information asymmetry on two levels:

By Client Nationality: AI algorithms process the scanned driving license prior to arrival. The system verifies whether the issuing country mandates an International Driving Permit (IDP), thereby preventing legal penalties.

Preventive Legal Education: AI notifies the client of mandatory equipment (reflective vest, warning triangle) and the sanctions under Article 291 of the Albanian Penal Code, which establishes that driving a vehicle while intoxicated or under the influence of narcotic substances, or without proof of the required competency, is punishable by imprisonment from ten days to three years.

5. Intelligent EV Management and Python Simulation

To address the challenge of electric vehicle (EV) management, the proposed model uses AI to illustrate how battery consumption might vary with terrain grade across Albanian topography. Below is an illustrative Python simulation, based on a simplified single-factor slope model, that "Rent a Car" businesses could adapt to give tourists an approximate estimate of predicted consumption:

```
import numpy as np

def calculate_ev_consumption(distance_km, average_slope_degrees,
    battery_capacity_kwh):
    """
    Illustrates approximate battery consumption of an EV based on topography and
    distance.
    The model applies a simplified baseline consumption of 0.15 kWh/km on flat roads,
    adjusted by a linear terrain factor. This is a conceptual demonstration, not a
    validated engineering model.
    """
    base_consumption_per_km = 0.15 # kWh/km

    # Terrain factor: gravity increases consumption uphill (slope > 0)
    if average_slope_degrees > 0:
        terrain_factor = 1 + (average_slope_degrees * 0.08)
    else:
```

```

        terrain_factor = 1 + (average_slope_degrees * 0.03) # Battery regeneration downhill

        total_consumption = distance_km * base_consumption_per_km * terrain_factor
        percentage_consumed = (total_consumption / battery_capacity_kwh) * 100

        return round(total_consumption, 2), round(percentage_consumed, 2)

# Simulation of two Albanian tourist routes for a 60 kWh EV
albania_routes = {
    "Tirana - Shkodra (Flat terrain)": {"distance": 100, "slope": 0.5},
    "Tirana - Saranda (via Llogara Pass - Mountainous terrain)": {"distance": 270,
"slope": 3.8}
}

print("--- ILLUSTRATIVE SIMULATION OF EV CONSUMPTION IN ALBANIA ---")
for route, data in albania_routes.items():
    kwh, percentage = calculate_ev_consumption(data["distance"], data["slope"],
battery_capacity_kwh=60)
    print(f"\nRoute: {route}")
    print(f"    -> Estimated consumption: {kwh} kWh")
    print(f"    -> Battery consumed: {percentage}% of total capacity (60 kWh)")

```

This simplified model applies a single linear slope-based multiplier and does not account for factors such as vehicle speed, ambient temperature, regenerative-braking efficiency limits, vehicle mass, or aerodynamic drag. Future work could validate and extend this framework against real-world EV telemetry data to produce a more precise, empirically grounded consumption model suitable for operational deployment.

5.1 Comparative Model: Internal Combustion Engine (ICE) Vehicle Consumption

To provide "Rent a Car" businesses with a comparative decision-support tool, the same illustrative approach can be extended to conventional gasoline-powered vehicles. The following simplified model estimates fuel consumption using a baseline liters-per-100km figure, adjusted by the same terrain-factor logic applied to the EV model above:

```
def calculate_ice_consumption(distance_km, average_slope_degrees, fuel_tank_liters):
```

```
    """
```

```

    Illustrates approximate fuel consumption of a gasoline (ICE) vehicle based on
    topography and distance. The model applies a simplified baseline consumption
    of 7.0 L/100km on flat roads, adjusted by a linear terrain factor. As with the
    EV model above, this is a conceptual demonstration, not a validated engineering
    model - it does not account for engine efficiency curves, gear ratios, traffic
    conditions, or driving behavior.
    """
```

```
    """
```

```
    base_consumption_per_100km = 7.0 # liters/100km
```

```
    # Terrain factor: gravity increases fuel consumption uphill (slope > 0)
```

```

# ICE vehicles do not meaningfully recover energy downhill, so the
# regenerative benefit is much smaller than for an EV
if average_slope_degrees > 0:
    terrain_factor = 1 + (average_slope_degrees * 0.10)
else:
    terrain_factor = 1 + (average_slope_degrees * 0.01) # Minimal downhill benefit

total_consumption_liters = (distance_km / 100) * base_consumption_per_100km * terrain_factor
percentage_of_tank = (total_consumption_liters / fuel_tank_liters) * 100

return round(total_consumption_liters, 2), round(percentage_of_tank, 2)

# Route data: distance in km, average slope in degrees (positive = net uphill)
albania_routes = {
    "Tirana to Saranda": {"distance": 270, "slope": 3.8},
    "Tirana to Shkodra": {"distance": 100, "slope": 0.5},
}

# Simulation of the same two Albanian tourist routes for a 50-liter fuel tank
print("--- ILLUSTRATIVE SIMULATION OF ICE VEHICLE CONSUMPTION IN ALBANIA ---")
for route, data in albania_routes.items():
    liters, percentage = calculate_ice_consumption(data["distance"], data["slope"], fuel_tank_liters=50)
    print(f"\nRoute: {route}")
    print(f" -> Estimated consumption: {liters} liters")
    print(f" -> Fuel tank consumed: {percentage}% of total capacity (50 L)")

```

The comparative output of these two models offers "Rent a Car" businesses a practical basis for customer-facing guidance and fleet-composition decisions. On the mountainous Tirana–Saranda route, the ICE model's steeper terrain penalty (a 0.10 multiplier versus the EV's 0.08) reflects the reality that internal combustion engines generally lose proportionally more efficiency on sustained inclines than electric drivetrains, which benefit from more efficient torque delivery at low speeds and partial energy recovery during descent. Conversely, on the flatter Tirana–Shkodra

route, the consumption gap between the two vehicle types narrows considerably, suggesting that terrain-aware trip planning may be more decision-relevant for EV range anxiety on mountainous routes than on coastal or lowland itineraries. For SME operators, this comparative framing could inform which vehicle type to recommend for a given route, or where to prioritize the deployment of charging infrastructure versus conventional fuel stations — reinforcing the paper's broader argument that AI-assisted, terrain-aware guidance systems add tangible value regardless of powertrain type, not solely for electric fleets.

6. Results and Discussion

The model above shows that on Albania's rugged terrain (such as the southern route), consumption increases by more than 30% compared to flat, linear roads, as a result of the gravitational force induced by the grade. When this AI algorithm is integrated into the Rent a Car application, it not only eliminates tourist anxiety but also suggests optimal charging stations in real time by connecting with the Ministry's platform.

7. Conclusions and Recommendations

This study confirms that the synergy between Artificial Intelligence and a centralized database from the Ministry of Tourism and Environment is a strategic necessity for Albania in 2026. This model protects small businesses (SMEs) from operational losses, increases legal compliance with the Penal Code (Article 291), and paves the way for the successful integration of green transport (EVs).

References (APA 7th Edition Style)

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